STUDY-UNI · UCC MARKING SCHEME

MA1100
Introductory Mathematics for Business I

Full Model Answers & Mark Allocation

Examination Session: Summer 2022 · **Paper:** 1

Module Code: MA1100 · **Duration:** 1.5 hours

Instructions to Candidates: Answer Question 1 and two other questions.

Prepared by: Study-Uni

Question 1

Total: 30 marks

(a) [5 marks]

Solve for n in the following: $3/8 = 12(1/2)^{n-1}$.

FULL MODEL ANSWER

Isolate the exponential term first, by dividing both sides by 12:

$$(1/2)^{n-1} = (3/8) / 12 = 3/96 = 1/32$$

Recognise 1/32 as a power of 1/2, since $2^5 = 32$:

$$(1/2)^{n-1} = (1/2)^5$$

Equating exponents (the bases match, so no logs are needed here):

$$n - 1 = 5$$

$$n = 6$$

Checknote: *Because both sides can be written as powers of the same base (1/2), this can be solved exactly by matching exponents — logarithms work too, but are unnecessary extra steps here.*

Mark Allocation — (a)

Tier	Marks	Criteria
Full credit	**5**	**Correctly isolates $(1/2)^{n-1} = 1/32$ AND recognises $1/32 = (1/2)^5$ (either directly or via logs) AND reaches n = 6.**
Good	3	Correct isolation of the exponential term, but solves via logarithms with a rounding slip instead of exact exponent matching, landing close to but not exactly n = 6.
Partial	2	Correctly isolates $(1/2)^{n-1} = 1/32$ but does not complete the solution for n.
Minimal	0	No valid isolation of the exponential term.

(b) [5 marks]

€200 is invested every month for 35 years at a rate of 3.6% p.a. compounded monthly. What is the value of the investment at the end of the 35 years?

FULL MODEL ANSWER

Use the future value annuity formula $V_n = A_0[(1+i)^n - 1] / i$, with the monthly deposit A_0 = €200, monthly rate i = 0.036/12 = 0.003, and n = 35 × 12 = 420 months:

$$V_{420} = 200 \times [(1.003)^{420} - 1] / 0.003$$

$$V_{420} = €167{,}918.53 \text{ (to the nearest cent)}$$

Mark Allocation — (b)

Tier	Marks	Criteria
Full credit	**5**	**Correctly identifies A_0 = 200, i = 0.003, n = 420 AND applies the future value annuity formula correctly AND reaches €167,918.53.**
Good	3	Correct formula and correct i, n, but a slip in the final evaluation.
Partial	2	Correct formula identified but incorrect conversion of the annual rate/term to monthly (e.g. using n = 35 instead of 420).
Minimal	0	Wrong formula used (e.g. present value formula, or simple compound interest with no annuity structure).

(c)(i) [1 marks]

Calculate the Annual Percentage Rate (APR) for an interest rate of 3.5% compounded yearly.

FULL MODEL ANSWER

$APR = (1 + r/m)^m - 1$. With $m = 1$ (compounded yearly), this reduces to $APR = r$ exactly:

$$APR = (1 + 0.035/1)^1 - 1 = 0.035$$

$$APR = 3.5000\%$$

Checknote: *When a nominal rate is already compounded annually, the APR equals the nominal rate exactly — no conversion is needed, since $m = 1$ makes $(1+r/m)^m - 1$ collapse to r.*

Mark Allocation — (c)(i)

Tier	Marks	Criteria
Full credit	**1**	**Recognises that a rate already compounded yearly IS its own APR (3.5%), with or without formally substituting m = 1.**
Minimal	0	Incorrect value given, or an unnecessary conversion applied that changes the figure.

(c)(ii) [2 marks]

Calculate the Annual Percentage Rate (APR) for an interest rate of 3.49% compounded monthly, to 4 decimal places.

FULL MODEL ANSWER

APR = $(1 + r/m)^m - 1$, with $r = 0.0349$, $m = 12$:

$$\text{APR} = (1 + 0.0349/12)^{12} - 1$$

$$\text{APR} = 3.5464\%$$

Mark Allocation — (c)(ii)

Tier	Marks	Criteria
Full credit	**2**	**Correct formula, correct substitution, AND APR = 3.5464% to 4dp.**
Minimal	0	Incorrect or missing formula.

(c)(iii) [2 marks]

Calculate the Annual Percentage Rate (APR) for an interest rate of 3.48% continuously compounded, to 4 decimal places.

FULL MODEL ANSWER

For continuous compounding, $APR = e^r - 1$, with $r = 0.0348$:

$$APR = e^{0.0348} - 1$$

$$APR = 3.5413\%$$

Checknote: *Note the three nominal rates in (c) are deliberately close (3.50%, 3.49%, 3.48%) but the resulting APRs are NOT in the same order as the nominal rates — compounding frequency matters as much as the headline rate. Always compute each on its own terms rather than assuming the ranking carries over.*

Mark Allocation — (c)(iii)

Tier	Marks	Criteria
Full credit	**2**	**Correct continuous-compounding formula AND APR = 3.5413% to 4dp.**
Minimal	0	Uses the discrete-compounding formula instead of e^r, or no coherent attempt.

(d)(i) [2 marks]

Find the derivative of $f(x) = (x^2 - 1)^{22}$.

FULL MODEL ANSWER

Apply the **chain rule**: outer function is $(\cdot)^{22}$, inner function is x^2-1:

$$f'(x) = 22(x^2-1)^{21} \times (2x)$$

$$f'(x) = 44x(x^2-1)^{21}$$

Mark Allocation — (d)(i)

Tier	Marks	Criteria
Full credit	**2**	**Correct chain rule application (outer power rule × inner derivative 2x) AND simplified to $44x(x^2-1)^{21}$.**
Partial	1	Correct outer power rule step, but inner derivative (2x) omitted or incorrect.
Minimal	0	Chain rule not applied (e.g. treats the bracket as if it were simply x).

(d)(ii) [2 marks]

Find the derivative of $f(x) = x^3e^{x^2}$.

FULL MODEL ANSWER

Apply the **product rule**, $u = x^3$, $v = e^{x^2}$ (itself needing the chain rule, since $v' = e^{x^2}\cdot 2x$):

$$f'(x) = 3x^2\cdot e^{x^2} + x^3\cdot(2x\cdot e^{x^2})$$

$$f'(x) = 3x^2e^{x^2} + 2x^4e^{x^2}$$

Factor out the common terms $x^2e^{x^2}$:

$$f'(x) = x^2e^{x^2}(3 + 2x^2)$$

Checknote: *A common slip is differentiating e^{x^2} as if it were e^x (forgetting the inner 2x from the chain rule) — always check the exponent itself is a function of x before treating $e^{g(x)}$ as a plain exponential.*

Mark Allocation — (d)(ii)

Tier	Marks	Criteria
Full credit	**2**	**Correct product rule AND correct chain rule on e^{x^2} AND factors to the simplest form $x^2e^{x^2}(3+2x^2)$.**
Partial	1	Correct product rule with the chain rule on e^{x^2} omitted (i.e. treats $d/dx(e^{x^2})$ as e^{x^2} rather than $2xe^{x^2}$).
Minimal	0	Product rule not applied.

(d)(iii) [2 marks]

Find the derivative of $f(x) = \ln(x) / e^x$.

FULL MODEL ANSWER

Apply the **quotient rule**, $u = \ln x$, $v = e^x$, so $u' = 1/x$, $v' = e^x$:

$$f'(x) = [e^x \cdot (1/x) - \ln(x) \cdot e^x] / (e^x)^2$$

Cancel a common factor of e^x between numerator and denominator:

$$f'(x) = [(1/x) - \ln(x)] / e^x$$

$$f'(x) = (1 - x \ln x) / (x e^x)$$

Mark Allocation — (d)(iii)

Tier	Marks	Criteria
Full credit	**2**	**Correct quotient rule AND correct simplification to $(1 - x \ln x)/(x e^x)$, or an equivalent fully simplified form.**
Partial	1	Correct quotient rule set-up but not simplified beyond the raw $[e^x/x - \ln(x)e^x]/e^{2x}$ form.
Minimal	0	Quotient rule not applied, or incorrect derivative of ln x or e^x.

(d)(iv) [2 marks]

Find the derivative of $f(x) = \sqrt{x^2 + 3}$.

FULL MODEL ANSWER

Write as a power and apply the **chain rule**:

$$f(x) = (x^2+3)^{1/2}$$

$$f'(x) = (1/2)(x^2+3)^{-1/2} \times 2x$$

$$f'(x) = x / \sqrt{x^2+3}$$

Mark Allocation — (d)(iv)

Tier	Marks	Criteria
Full credit	**2**	**Correct chain rule application AND simplified to $x/\sqrt{x^2+3}$.**
Partial	1	Correct power/chain rule set-up with the factor of 2 from the inner derivative not cancelled correctly.
Minimal	0	Chain rule not applied (inner derivative 2x omitted).

(e) [7 marks]

Given that $f(x) = x^3 - 4x^2 - 3x + 1$, find $f'(x)$ and $f''(x)$. Hence find the coordinates of the local maximum and the local minimum of f.

FULL MODEL ANSWER

Differentiate once, then again:

$$f'(x) = 3x^2 - 8x - 3$$

$$f''(x) = 6x - 8$$

Find stationary points by setting $f'(x) = 0$:

$$3x^2 - 8x - 3 = 0$$

This factorises (or solve via the quadratic formula) to give:

$$x = 3 \quad \text{or} \quad x = -1/3$$

Classify using the second derivative:

x	f″(x)	Nature	f(x)
$x = -1/3$	$6(-1/3)-8 = -10 < 0$	Local maximum	41/27
$x = 3$	$6(3)-8 = 10 > 0$	Local minimum	−17

$$\text{Local maximum at } (-1/3, 41/27)\text{; local minimum at } (3, -17)$$

Checknote: *41/27 is an easy value to mis-simplify — keep the fraction exact (−1/27 − 4/9 + 1 + 1) rather than rounding early, since 4/9 must first be converted to twenty-sevenths (12/27) to combine correctly.*

Mark Allocation — (e)

Tier	Marks	Criteria
Full credit	**7**	**Correct f′(x) and f″(x) AND correctly solves f′(x)=0 for both x=3 and x=−1/3 AND correctly classifies each using f″(x) AND correctly computes both y-coordinates.**
Good	5	Correct f′(x), f″(x), and both x-values with correct classification, but an arithmetic slip in one of the two y-coordinates.
Partial	3	Correct f′(x) and both stationary x-values found, but f″(x) not used to classify them (or classification incorrect).
Minimal	1	f′(x) correct but stationary points not correctly solved for.

Question 1 — Total 30/30 marks confirmed

Question 2

Total: 25 marks

(a) [8 marks]

The supply function for an item is $P_s = 2Q + 22$ and the demand function is $P_d = 50 - 1.5Q$. Find the equilibrium value of Q for the production of the items and the corresponding unit price per item.

FULL MODEL ANSWER

At equilibrium, $P_s = P_d$:

$$2Q + 22 = 50 - 1.5Q$$

$$3.5Q = 28 \;\Rightarrow\; Q = 8$$

Substitute back into either equation to find P:

$$P = 2(8) + 22 = 16 + 22$$

EQUILIBRIUM

$$Q^* = 8,\;\; P^* = €38$$

Mark Allocation — (a)

Tier	Marks	Criteria
Full credit	**8**	**Correctly sets $P_s = P_d$, solves for Q = 8 AND P = €38, with both values clearly stated.**
Good	6	Correct method and correct Q, with an arithmetic slip carried into the final P value (or vice versa).
Partial	3	Correct equation set up ($P_s = P_d$) but not solved through to numeric values.
Minimal	0	Incorrect equilibrium condition used.

(b) [9 marks]

If a tax of €10.50 is levied on the sale of each item, what will the new equilibrium price and quantity be?

FULL MODEL ANSWER

A per-unit tax shifts the supply curve **up** by the amount of the tax:

$$P_{s(\text{new})} = 2Q + 22 + 10.5 = 2Q + 32.5$$

Set the new supply curve equal to the (unchanged) demand curve:

$$2Q + 32.5 = 50 - 1.5Q$$

$$3.5Q = 17.5 \;\Rightarrow\; Q = 5$$

Substitute into the demand curve to get the price consumers pay:

$$P = 50 - 1.5(5) = 50 - 7.5 = 42.5$$

$$\text{New } Q^* = 5,\ \text{new price paid by consumers} = €42.50$$

The price producers keep net of tax: €42.50 − €10.50 = **€32.00** — needed for part (c).

Mark Allocation — (b)

Tier	Marks	Criteria
Full credit	**9**	**Correctly shifts the supply curve up by €10.50 AND solves the new equilibrium AND correctly reports Q = 5, consumer price = €42.50 (with producer's net price identified as €32.00).**
Good	6	Correct direction of shift and correct new equilibrium Q and consumer price, but the distinction between consumer price and producer's net price not drawn out.
Partial	3	Correctly identifies that supply shifts by €10.50 but shifts it the wrong direction, carrying through consistently.
Minimal	0	Tax applied to the demand curve instead of supply, or no shift applied at all.

(c) [8 marks]

In what ratio will the burden of this tax be shared between the producer and the consumer?

FULL MODEL ANSWER

Compare each side's price before and after the tax (using the original equilibrium price €38 from part (a) as the baseline):

Party	Price before	Price after	Burden
Consumer pays	€38.00	€42.50	**€4.50**
Producer receives (net)	€38.00	€32.00	**€6.00**
Total (must equal the €10.50 tax)			**€10.50**

Expressing the burdens as a ratio, producer : consumer:

$$6.00 : 4.50 \;\equiv\; 4 : 3$$

The producer bears the larger share here, since supply is the steeper (less elastic) curve — slope 2 in magnitude versus demand's slope of 1.5. The steeper (less elastic) side of the market is less able to adjust quantity in response to the tax, and so absorbs more of its burden.

> **Checknote:** *Always re-derive the incidence from the actual slopes given in each question rather than assuming a 'usual' direction — here supply is the steeper (less elastic) curve, so the producer, not the consumer, carries the larger share.*

Mark Allocation — (c)

Tier	Marks	Criteria
Full credit	**8**	**Correctly computes both the consumer price rise (€4.50) and producer price fall (€6.00), AND confirms they sum to €10.50, AND states the ratio 4:3 (producer:consumer).**
Good	6	Correct ratio 4:3 obtained, without explicit reference to the sum-to-€10.50 check or the elasticity explanation.
Partial	3	One of the two burden amounts correctly computed, but the ratio not correctly formed.
Minimal	0	No coherent attempt to split the €10.50 tax between the two parties.

Question 2 — Total 25/25 marks confirmed

Question 3

Total: 25 marks

(a) [8 marks]

A couple decide they want to purchase a house valued at €215,000. They decide to save €1,000 each month for 1.5 years for a deposit. If the bank offers a rate of 3.6% p.a. compounded monthly, how much will they have saved in this time?

FULL MODEL ANSWER

Use the future value annuity formula V_n = $A_0[(1+i)^n - 1] / i$, with the monthly deposit A_0 = €1,000, monthly rate i = 0.036/12 = 0.003, and n = 1.5 × 12 = 18 months:

$$V_{18} = 1\{,\}000 \times [(1.003)^{18} - 1] / 0.003$$

$$V_{18} = €18{,}466.43 \text{ (to the nearest cent)}$$

Checknote: *The €215,000 house value given in the preamble is not needed for this part — it's context/set-up for the mortgage in part (b). Watch for scene-setting figures that aren't inputs to the specific sub-part being asked.*

Mark Allocation — (a)

Tier	Marks	Criteria
Full credit	**8**	**Correctly identifies A_0 = 1,000, i = 0.003, n = 18 AND applies the future value annuity formula correctly AND reaches €18,466.43.**
Good	6	Correct formula and correct i, n, but a slip in the final evaluation.
Partial	4	Correct formula identified but incorrect conversion of the time period to months (e.g. using n = 1.5 instead of 18).
Minimal	1	Wrong formula used, or no coherent attempt.

(b) [8 marks]

> *The couple are offered a mortgage of €200,000 at 4.8% p.a. compounded monthly. If they now can afford to pay €1,400 a month, how long, to the nearest year, will it take to pay off the mortgage?*

FULL MODEL ANSWER

Use the present value annuity formula $V_0 = A_0[1 - (1+i)^{-n}] / i$, with V_0 = €200,000, A_0 = €1,400, $i = 0.048/12 = 0.004$. Solve for n:

$$200\{,\}000 = 1\{,\}400 \times [1 - (1.004)^{-n}] / 0.004$$

$$1 - (1.004)^{-n} = 200\{,\}000 \times 0.004 / 1\{,\}400 = 0.57143$$

$$(1.004)^{-n} = 0.42857$$

Take logs of both sides and solve for n:

$$-n \cdot \ln(1.004) = \ln(0.42857) \;\Rightarrow\; n = 212.25 \text{ months}$$

$$212.25 \text{ months} = 17.69 \text{ years}$$

$$\approx 18 \text{ years (to the nearest year)}$$

> **Checknote:** *17.69 years rounds UP to 18 years to the nearest year, not down to 17 — a common slip is truncating rather than rounding, which would understate how long the mortgage actually takes.*

Mark Allocation — (b)

Tier	Marks	Criteria
Full credit	**8**	**Correctly sets up the present value annuity formula with the given values AND correctly isolates $(1.004)^{-n}$ AND solves via logs to n ≈ 17.69 years AND correctly rounds to 18 years.**
Good	6	Correct set-up and correct isolation of the exponential term, with a slip in the final log-solving step or in the final rounding.
Partial	4	Correct formula identified and correctly substituted, but not successfully rearranged to isolate the exponential term.
Minimal	1	Uses the future value formula instead of present value, or does not attempt to solve for n via logarithms.

(c) [9 marks]

Later they decide to sell the house. There are 10 years of payments remaining on the mortgage. How much must they repay the bank to pay off their mortgage?

FULL MODEL ANSWER

The amount required to clear the remaining mortgage is the present value of the 10 years of remaining monthly payments, at the same terms (A_0 = €1,400, i = 0.004). With 10 years remaining, n = 10 × 12 = 120 months:

$$V_{0(\text{remaining})} = 1\{,\}400 \times [1 - (1.004)^{-120}] / 0.004$$

$$V_{0(\text{remaining})} = €133{,}218.36 \text{ (to the nearest cent)}$$

Checknote: *This is a present value annuity calculation on the REMAINING term only (10 years = 120 months), not the original mortgage term from part (b) — the balance owed depends on payments still to come, not the full original schedule.*

Mark Allocation — (c)

Tier	Marks	Criteria
Full credit	**9**	**Correctly recognises the repayment amount as the present value of the remaining 120 monthly payments AND correctly substitutes A_0 = 1,400, i = 0.004, n = 120 AND reaches €133,218.36.**
Good	7	Correct formula and correct substitution, with a rounding or calculator slip in the final value.
Partial	4	Correct present value formula identified but incorrect n used (e.g. total mortgage term rather than the 10 years remaining).
Minimal	1	Approach does not use the annuity present value structure at all (e.g. simple multiplication of 1,400 × 120).

Question 3 — Total 25/25 marks confirmed

Question 4

Total: 25 marks

4(a)(i) [4 marks]

A supply function is given by $Q = 900 - P^2$. Calculate the elasticity equation as a function of P.

FULL MODEL ANSWER

Price elasticity of supply is E = (dQ/dP) × (P/Q). First differentiate Q with respect to P:

$$dQ/dP = -2P$$

Then form the elasticity expression using the original Q(P):

$$E(P) = -2P \times P / (900 - P^2) = -2P^2 / (900 - P^2)$$

Mark Allocation — 4(a)(i)

Tier	Marks	Criteria
Full credit	**4**	**Correctly differentiates Q to get dQ/dP = −2P AND correctly forms E(P) = (dQ/dP)(P/Q) with the full expression for Q in the denominator.**
Partial	2	Correct dQ/dP but elasticity formula assembled incorrectly (e.g. Q/P used instead of P/Q).
Minimal	0	Incorrect differentiation of Q with respect to P.

4(a)(ii) [2 marks]

Find the price elasticity of supply when the current price is P = 15.

FULL MODEL ANSWER

Substitute P = 15 into the elasticity expression from (a)(i):

$$E(15) = -2(15)^2 / (900 - 15^2) = -450 / 675$$

$$E(15) = -2/3 \approx -0.67$$

Checknote: *This particular supply function actually has negative price elasticity of supply for P < 30 (unusual for a real-world 'supply' curve) — follow the algebra given in the question exactly rather than adjusting the sign to fit intuition.*

Mark Allocation — 4(a)(ii)

Tier	Marks	Criteria
Full credit	**2**	**Correctly substitutes P = 15 into the student's own E(P) from (a)(i) AND reaches E(15) = −2/3.**
Minimal	0	Substitution not attempted or incorrect.

4(a)(iii) [7 marks]

Find the price that will maximise the revenue.

FULL MODEL ANSWER

Revenue is price times quantity, using the given Q(P):

$$R(P) = P \times Q = P(900 - P^2) = 900P - P^3$$

Maximise by setting the first derivative to zero:

$$dR/dP = 900 - 3P^2 = 0 \;\Rightarrow\; P^2 = 300$$

$$P = \pm\sqrt{300} = \pm 10\sqrt{3}$$

Reject the negative root, since price cannot be negative:

$$P = 10\sqrt{3} \approx 17.32$$

Confirm it is a maximum using the second derivative:

$$d^2R/dP^2 = -6P$$

$$\text{At } P = 10\sqrt{3}\text{: } d^2R/dP^2 = -60\sqrt{3} < 0$$

CONFIRMED MAXIMUM

$$P^* = 10\sqrt{3} \approx €17.32$$

Mark Allocation — 4(a)(iii)

Tier	Marks	Criteria
Full credit	**7**	**Correctly forms $R(P) = 900P - P^3$ AND sets $dR/dP = 0$ to get $P^2 = 300$ AND correctly rejects the negative root AND verifies via the second derivative that $P = 10\sqrt{3}$ is a maximum.**
Good	5	Correct revenue function and correct $P = 10\sqrt{3}$ found, but the second-derivative maximum check is missing.
Partial	3	Correct revenue function formed and differentiated, but the negative root not explicitly rejected.
Minimal	1	Revenue not correctly formed as $P \times Q$.

4(b)(i) [3 marks]

A company sells Q items at a cost of P per item. The demand function for the items is given by $P = 300 - 0.02Q$. Find the revenue function in terms of Q.

FULL MODEL ANSWER

Revenue is price times quantity, using the demand function to express price in terms of Q:

$$R(Q) = P \times Q = (300 - 0.02Q)Q$$

$$R(Q) = 300Q - 0.02Q^2$$

Mark Allocation — 4(b)(i)

Tier	Marks	Criteria
Full credit	**3**	**Correctly substitutes the demand function into $R = P \times Q$ AND expands to $300Q - 0.02Q^2$.**
Partial	1	Correct structure $R = P \times Q$ identified but not expanded, or an expansion slip.
Minimal	0	Revenue not correctly formed as price times quantity.

4(b)(ii) [3 marks]

> *There are fixed costs of €9,000 per year and each item costs €30 to manufacture. Find the cost function in terms of Q.*

FULL MODEL ANSWER

Total cost is fixed costs plus total variable cost. Here the €30 is already a per-unit cost, so total variable cost is simply 30Q:

$$C(Q) = 9\{,\}000 + 30Q$$

Mark Allocation — 4(b)(ii)

Tier	Marks	Criteria
Full credit	**3**	**Correctly identifies fixed costs (9,000) AND variable cost as 30 per unit AND combines to C(Q) = 9,000 + 30Q.**
Partial	1	One component correct (fixed or variable) but not combined correctly.
Minimal	0	Cost function not correctly formed.

4(b)(iii) [6 marks]

Find the production quantity Q for which the profit is maximised. Find the corresponding unit price.

FULL MODEL ANSWER

Profit is revenue minus cost, using the functions from (b)(i) and (b)(ii):

$$\pi(Q) = (300Q - 0.02Q^2) - (9\{,\}000 + 30Q)$$

$$\pi(Q) = -0.02Q^2 + 270Q - 9\{,\}000$$

Maximise by setting the first derivative to zero:

$$d\pi/dQ = -0.04Q + 270 = 0 \;\Rightarrow\; Q = 6\{,\}750$$

Check the second derivative: $d^2\pi/dQ^2 = -0.04 < 0$, confirming a **maximum**.

CONFIRMED MAXIMUM

Corresponding price, from the demand function:

$$P = 300 - 0.02(6\{,\}750) = 300 - 135$$

$$Q^* = 6{,}750 \text{ units},\;\; P^* = €165$$

Mark Allocation — 4(b)(iii)

Tier	Marks	Criteria
Full credit	**6**	**Correctly forms profit as R(Q) − C(Q) using the student's own (b)(i) and (b)(ii) AND sets dπ/dQ = 0 to get Q = 6,750 AND verifies it is a maximum AND correctly finds P = €165.**
Good	4	Correct profit function and correct Q = 6,750, but the second-derivative maximum check or the final price is missing.
Partial	2	Profit function correctly differentiated, but an arithmetic slip in solving for Q.
Minimal	0	Profit not correctly formed as revenue minus cost.

Question 4 — Total 25/25 marks confirmed